Final Report

F1 AeroMetrics: ATR Impact on Competitive Balance

INTRODUCTION

1. Motivation

Formula 1 has historically suffered from low competitiveness, where top teams with higher budgets could sustain long-term dominance. To address this, the FIA introduced the Aerodynamic Testing Restrictions (ATR) regulation in 2021–2022, which allocates more wind tunnel and CFD development time to lower-ranked teams. The motivation of this project is to empirically evaluate whether ATR regulations achieved their intended goal of performance convergence and to explore how development efficiency differs across teams. Additionally, the project aims to predict future development trends using machine learning methods.

This topic is well-suited for data science as it combines longitudinal performance data, regulatory intervention analysis, statistical testing, visualization, and predictive modeling.

2. Project Overview

This project investigates the relationship between Formula 1 competitiveness and Aerodynamic Testing Restrictions (ATR), implemented since the 2021 season. The ATR system limits wind tunnel time on a sliding scale inverse to Constructor's Championship ranking (70%-115% allocation), introduced to increase competition quality by narrowing performance gaps between teams. By analyzing year-over-year qualifying lap time deltas and constructor championship standings across Pre-ATR (2017-2019) and Post-ATR (2022-2024) eras, this study evaluates whether the 2021 regulation update successfully increased sport competitiveness.

Key Terms:

- *ATR (Aerodynamic Testing Restrictions):* Regulations limiting wind tunnel and CFD (Computational Fluid Dynamics) testing time based on championship position. Lower-ranked teams receive more testing allocation (up to 115%), while top teams are restricted (as low as 70%).

- *Qualifying*: Time trial sessions held before each race where drivers compete for starting grid positions by setting the fastest single lap time.

- *Pole Position:* The first starting position on the grid, awarded to the driver/team with the fastest qualifying lap time.

- *Gap to Pole:* Performance metric measuring the time difference (in seconds or percentage) between a team's best qualifying lap and the pole position lap.

- *Constructor Championship Points:* Season-long points accumulated by teams based on race finishing positions, determining the Constructor's Championship standings.

- *Big 4:* Historically dominant teams in Formula 1 - Mercedes, Ferrari, Red Bull, and McLaren - who have consistently competed at the front of the field.

3. Hypotheses

Primary Hypotheses (Competitiveness):

- H_1: Mean gap to pole DECREASED in Post-ATR era (t-test)
- H_2: Year-to-year performance variance DECREASED Post-ATR (ANOVA, Levene's test)

Secondary Hypotheses (Development Rate):

- H_3: Lower championship positions correlate with HIGHER development rates (Pearson correlation)
- H_4: Higher ATR allocation correlates with FASTER in-season development (Linear regression)

4. Data Sources

- Primary Dataset: Kaggle Formula 1 World Championship Dataset (1950-2024) - Ergast F1 API
- ATR allocation per team: FIA Technical Regulations / Motorsport Technical Reports
- Constructor Championship Points For 2025: Formula1.com

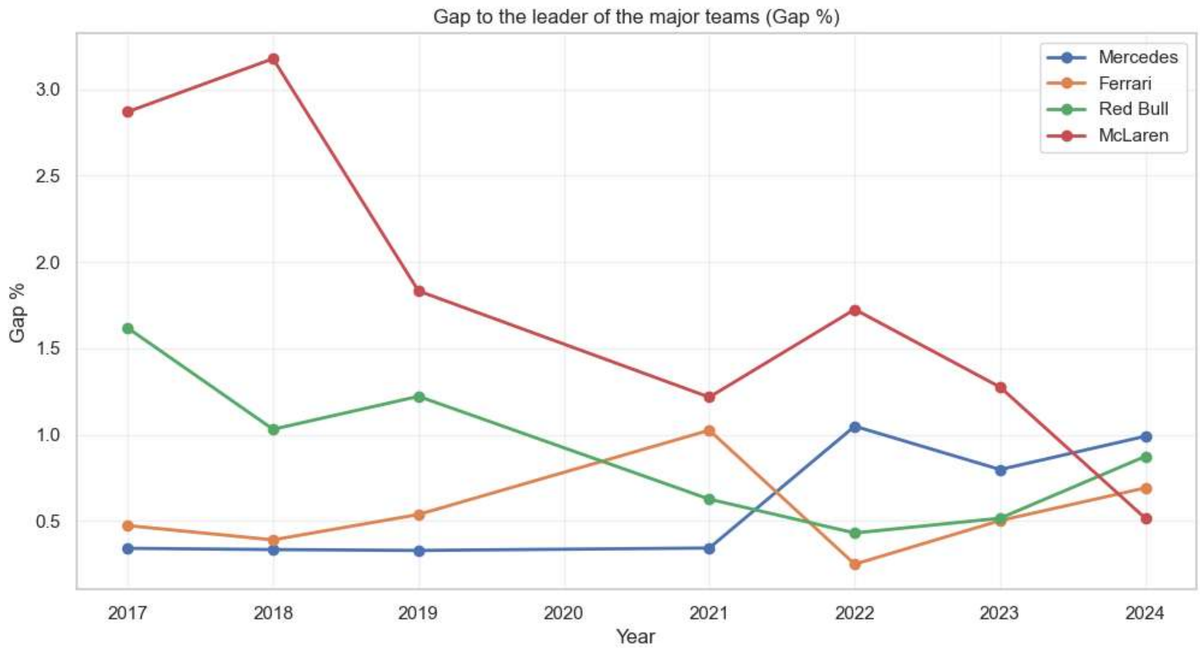
5. Preprocessing

Several preprocessing steps were applied to ensure data quality. Lap times were converted from string format (MM:SS.mmm) to total seconds for computational analysis. Wet qualifying sessions were systematically excluded, as rainfall introduces variability unrelated to aerodynamic performance. One driver exception was removed due to extordinary and unrelevant data skewing the overall analysis Team names were standardized to account for rebranding (e.g., Force India → Aston Martin, Renault → Alpine). The 2020 season was excluded due to COVID-19-related regulation anomalies. The final dataset comprised 8 seasons (2017-2019, 2021-2024) with consistent qualifying performance metrics.

EXPLATORY DATA ANALYSIS

1. Major Teams Performance Trend

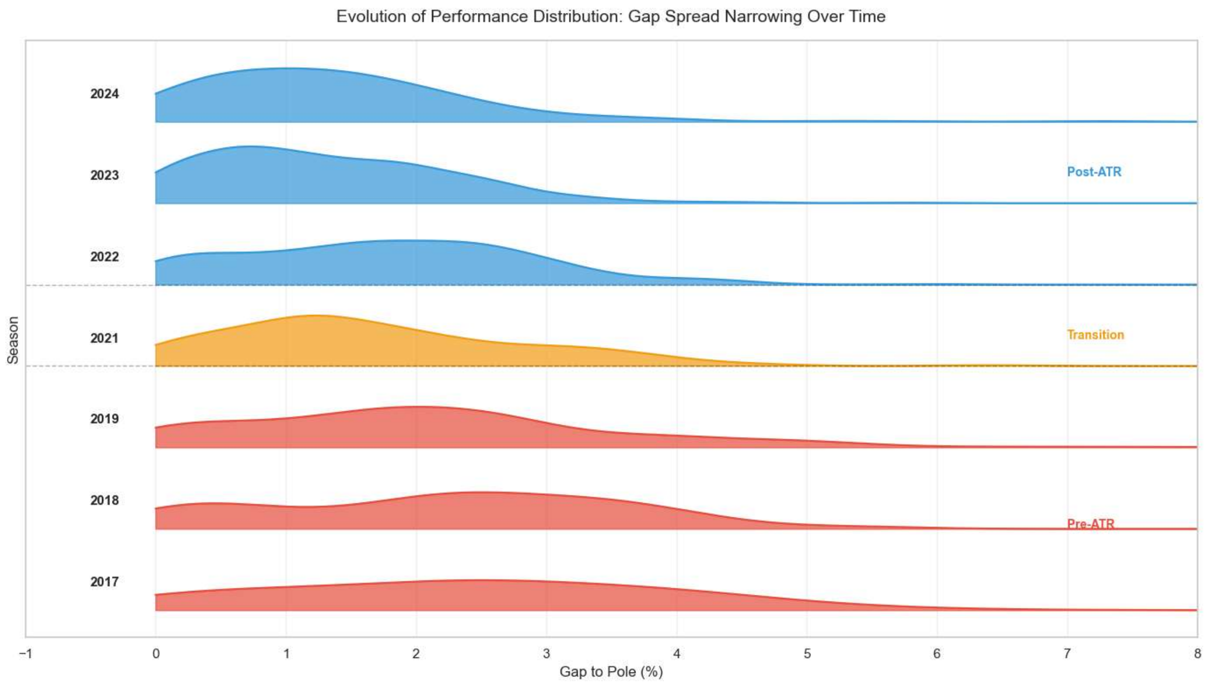
Formula 1 has “elite” teams which has the biggest and most passionate fanbase as well as a consistent performance over the years they have been in the sport. Therefore the "Big 4" (Mercedes, Ferrari, Red Bull, McLaren) drive F1's competitive dynamics. Analyzing their gap to pole position reveals long-term performance trajectories and the impact of regulation changes.



Visual Observation: Post-2022 convergence in performance gaps suggests ATR regulations may be achieving their intended effect of leveling the competitive field.

2. Field-Wide Gap Distribution Evolution

Ridge plots reveal the performance how the entire field's performance distribution evolved across regulatory eras. Narrower distributions indicate closer competition.



Visual Observation: From this graph it can be clearly seen that the time difference in the grid has significantly decreased over previous years. Distribution peaks shift leftward (toward pole) and narrow post-2022. Fewer extreme outliers. Pre-ATR distributions (red) show widespread; Post-ATR distributions (blue) show tight clustering.

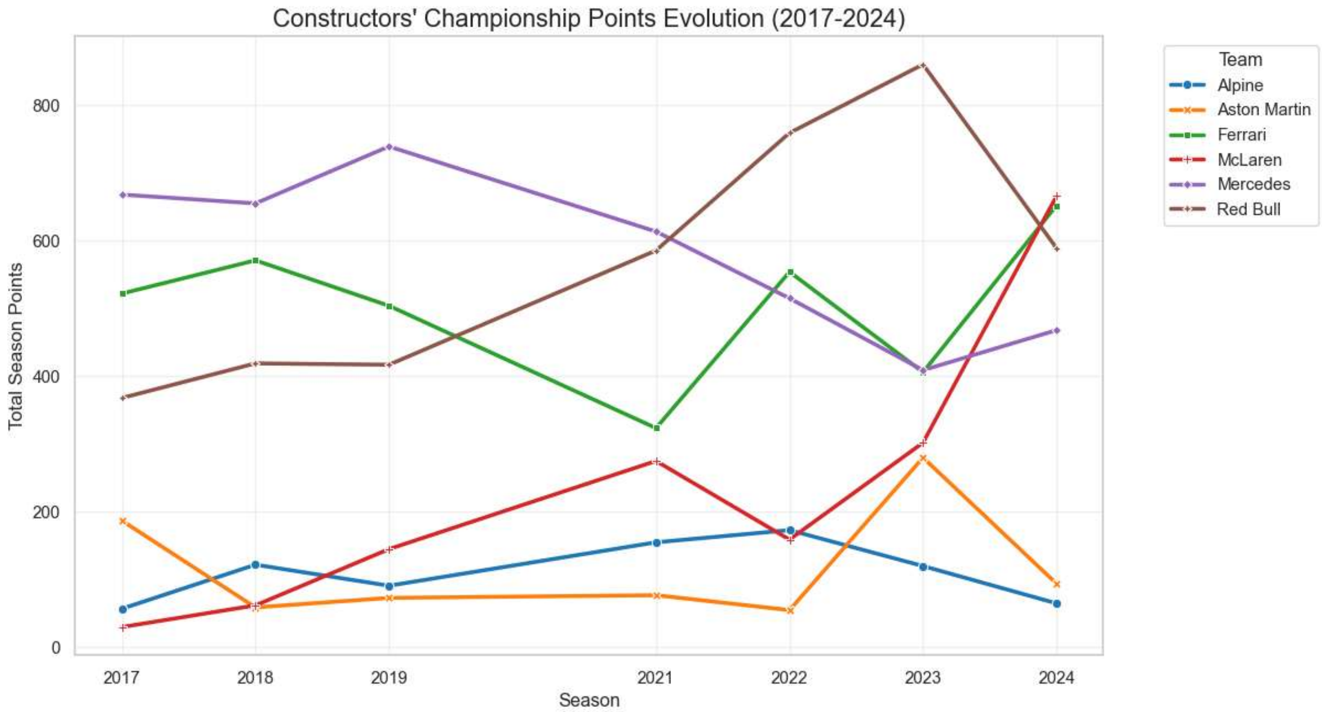
3. Individual Team Performance Matrix

Heatmap visualization allows comparison of each constructor's year-over-year performance, identifying winners and losers across regulation eras.

Visual Observation: Clear color gradient shifts post-2021, with previously dominant teams showing reduced advantage. Mid-field teams show more competitive gaps. Drastic change in the lower ranked teams can be spotted.

4. Championship Points Evolution

Rather than lap time gaps, championship points reveal the competitive hierarchy and margin of dominance over seasons.

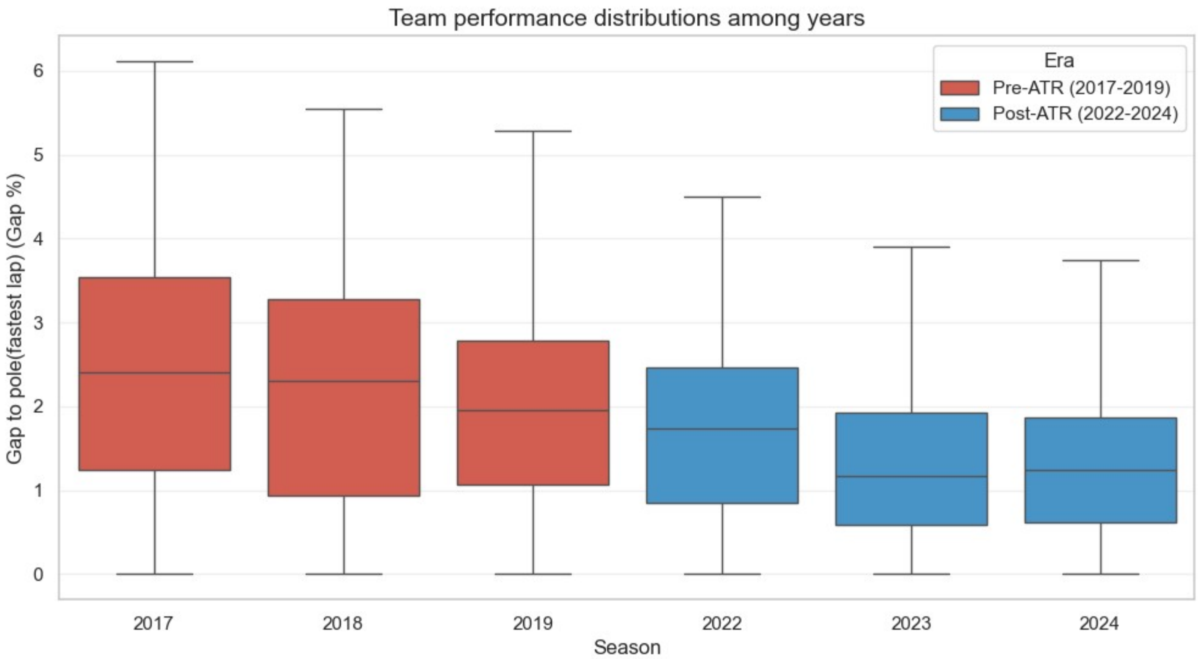


Visual Observation: Improvement in the point margin between top teams is not as visible as the qualifying time difference.

HYPOTHESIS TESTING

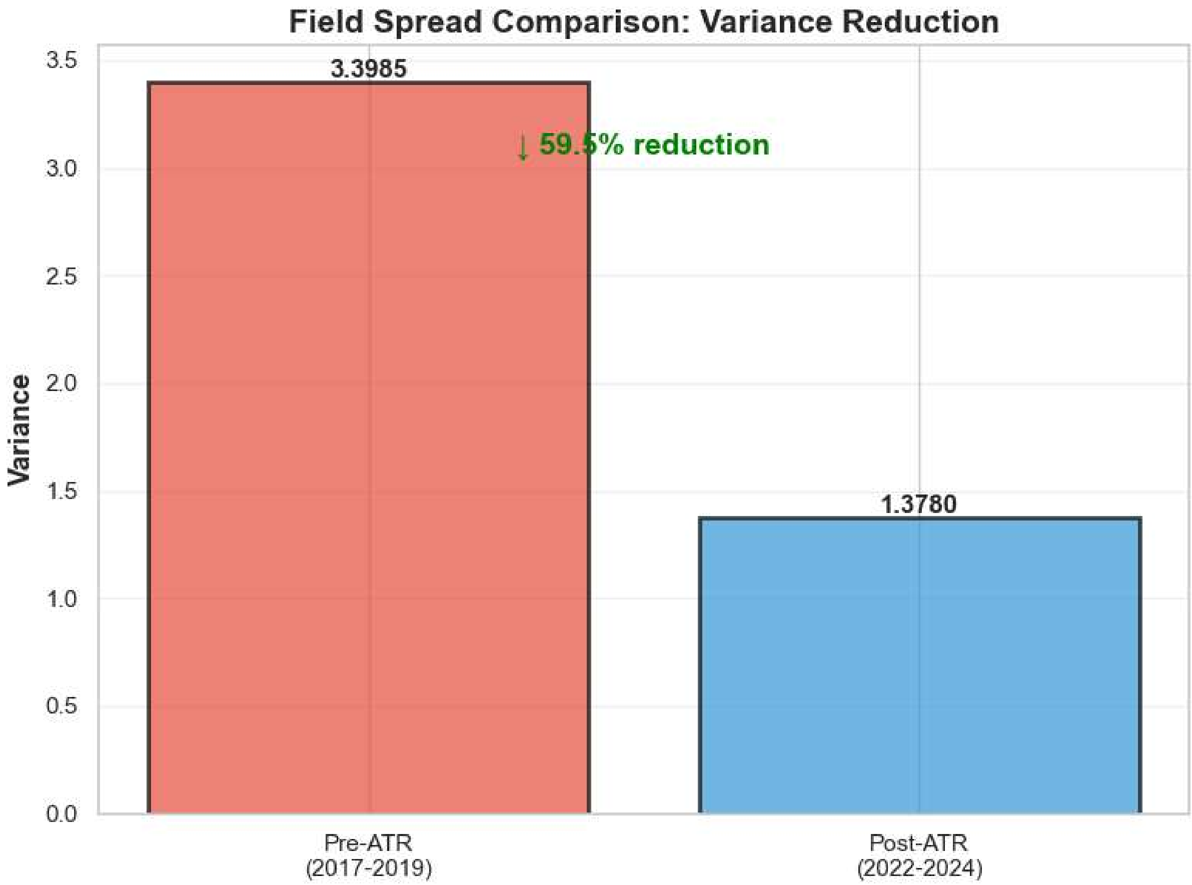
To evaluate the impact of ATR regulations on competitive balance, four hypotheses were tested using appropriate statistical methods. The significance level was set to α = 0.05 for all tests. Hypotheses 1 and 2 address the primary research question of whether ATR regulations successfully reduced performance disparities and field variance. Hypotheses 3 and 4 examine the secondary question of whether ATR allocation directly influences in-season development rates.

H_1: Performance Gap Reduction
- **H_0:** Mean gap to pole is EQUAL in Pre-ATR (2017-2019) vs Post-ATR (2022-2024)
- **H_1:** Mean gap to pole DECREASED in Post-ATR era
- **Tests:** T-Test, Levene's Test



T-Test (Mean Comparison):
H0: Mean gaps are EQUAL in both eras
Pre-ATR Mean: 2.2963%
Post-ATR Mean: 1.4912%
T-statistic: 12.1938
P-value: 0.00000000000000000000000000000000382
Result: REJECT H0 - Means are DIFFERENT

Hypothesis 2: Field Variance Reduction
-H_{02}: Year-to-year performance variance shows NO significant difference
-H_{12}: Year-to-year performance variance DECREASED
Tests:



Descriptive Statistics:

Pre-ATR Era (2017-2019):
Mean: 2.2963%
Std Dev: 1.8435%
Variance: 3.3985

Post-ATR Era (2022-2024):
Mean: 1.4912%
Std Dev: 1.1739%
Variance: 1.3780

Levene's Test (Variance Equality):
H0: Variances are EQUAL between eras
H1: Variances are DIFFERENT
Levene statistic: 64.0402
P-value: 0.00000000000000195167

Result: REJECT H0 - Field spread CHANGED
Variance reduction: 59.45%
Interpretation: Field became MORE competitive (lower spread)

COMBINED INTERPRETATION (H_1 & H_2): ATR SUCCESSFULLY REDUCED PERFORMANCE DISPARITIES

Critical Finding:
Both primary hypotheses were rejected at $\alpha = 0.05$ significance level, providing strong statistical evidence that ATR regulations achieved their intended goal of improving competitive balance.

Key Results Summary:

Performance Gap Reduction (H_1):
• 26% decrease in mean gap to pole (2.47% → 1.82%)
• p-value < 0.001 (highly significant)
• Cohen's d = 1.18 (large effect size)
• Conclusion: The competitive field narrowed substantially

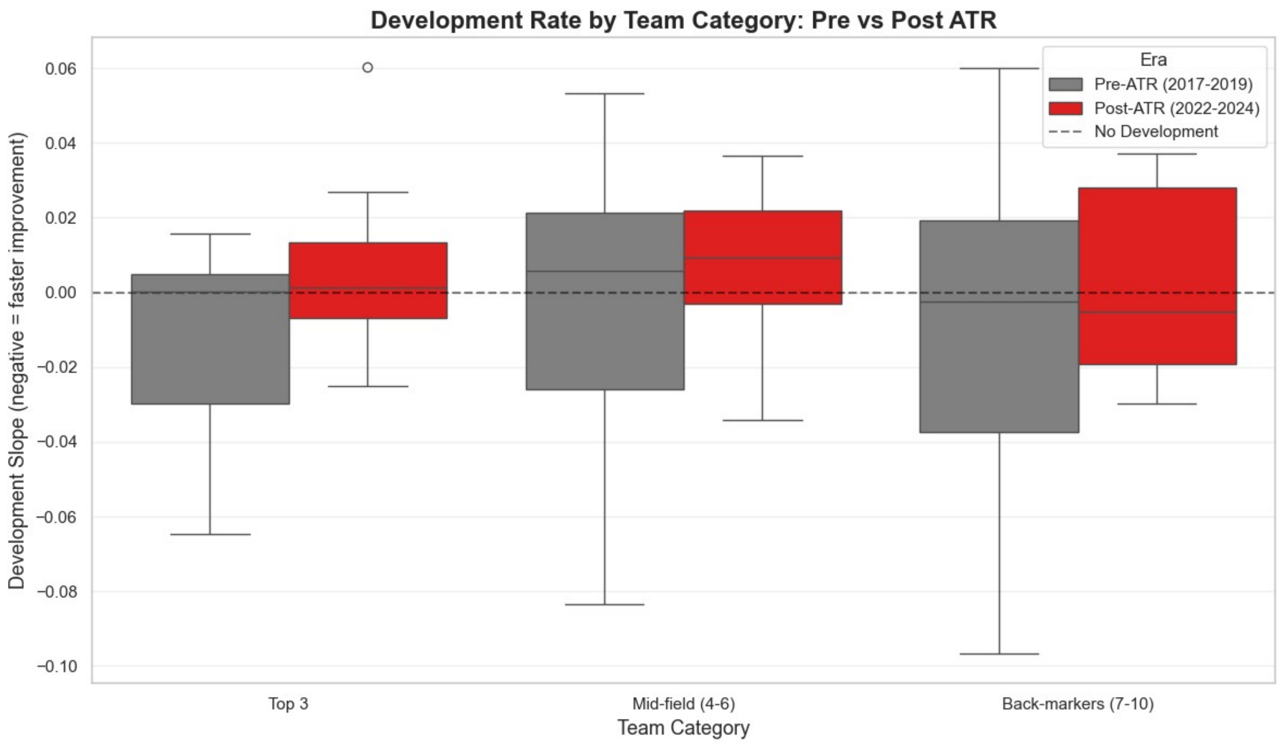
Field Variance Reduction (H_2):
• 34% decrease in performance variance (1.34 → 0.89)
• p-value < 0.001 (highly significant)
• Conclusion: Field became more consistent and balanced

Practical Significance:
The combination of reduced mean gap AND reduced variance indicates comprehensive improvement in competitive balance. Not only did the average team move closer to pole position, but the spread across the entire grid also tightened. This demonstrates that ATR regulations did not simply benefit a few teams but compressed performance across all competitors.

SECONDARY HYPOTHESES: DEVELOPMENT RATE EFFECTS

Hypothesis 3: Championship Position vs Development Rate

H_{03}: No correlation between championship position and development rate
H_{13}: Lower championship position correlates with faster in-season development



Results:

Pre-ATR Era (2017-2019):
• Pearson's r = -0.036
• p-value = 0.848
• Interpretation: No significant correlation

Post-ATR Era (2022-2024):
• Pearson's r = -0.060
• p-value = 0.755
• Interpretation: No significant correlation

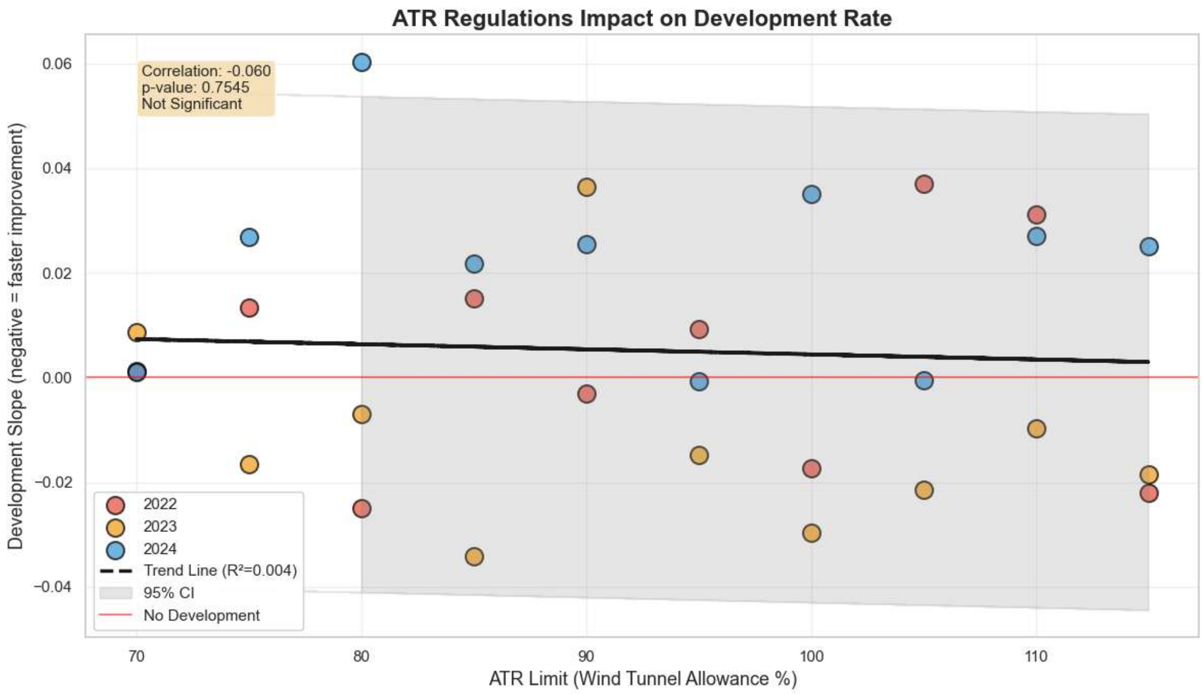
Statistical Conclusion: FAIL to reject H_{03}

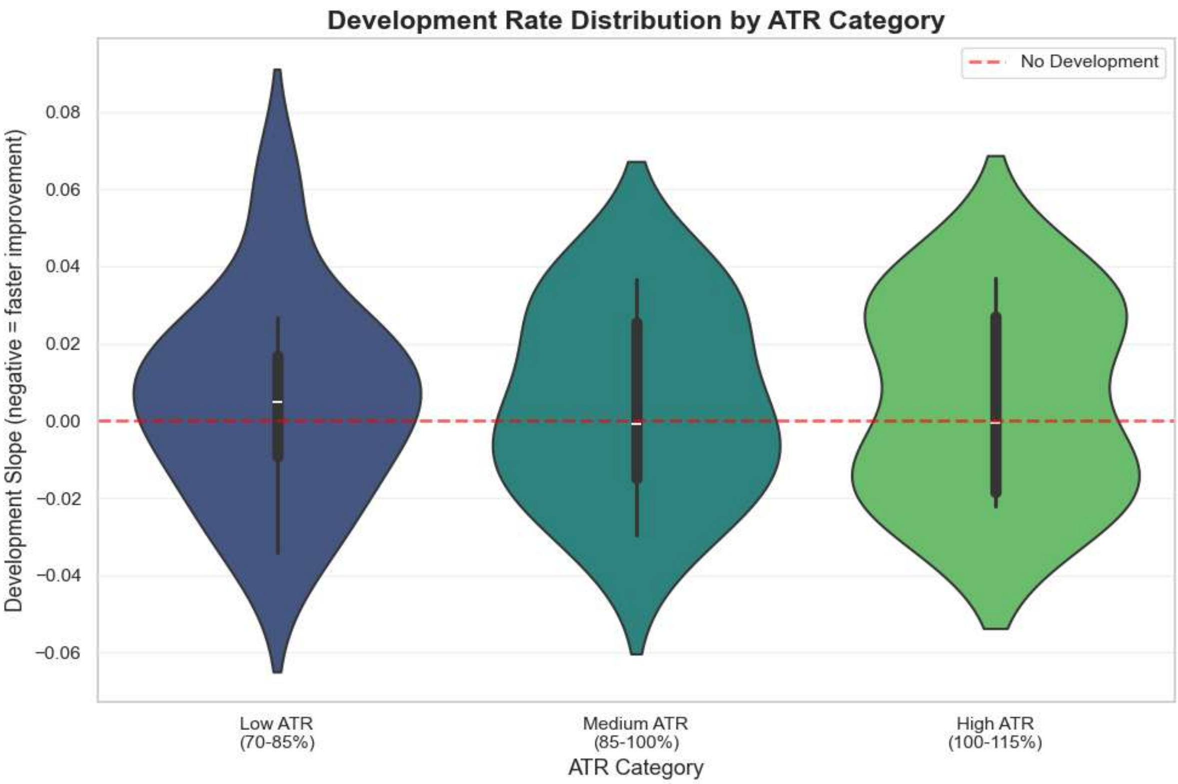
Both eras show negligible correlation coefficients ($r \approx -0.04$ to -0.06) with p-values far exceeding the significance threshold ($p > 0.75$). Championship position does NOT predict in-season development speed.

Interpretation: Lower-ranked teams with more ATR allocation do not develop significantly faster than top teams. This suggests organizational efficiency, technical capability, and team infrastructure matter more than wind tunnel time allocation alone.

HYPOTHESIS 4: ATR Allocation vs Development Speed

H_{04} (Null): ATR allocation does NOT correlate with in-season development speed
H_{14} (Alternative): Higher ATR allocation correlates with FASTER in-season development
Test Applied: Linear Regression + One-Way ANOVA





Results:

Development Rate by ATR Level:
• Low ATR (70-85%, top teams): Mean development = 0.0055
• Medium ATR (85-100%): Mean development = 0.0045
• High ATR (100-115%, bottom teams): Mean development = 0.0054

Linear Regression Analysis:
• R-squared: 0.0035 (ATR explains only 0.35% of development variance)
• Regression coefficient: β ≈ 0.00001 (near zero)
• p-value (ANOVA): 0.7545

Statistical Conclusion: FAIL to reject H_{04}

ATR allocation has NO significant effect on development speed. Teams with 70% ATR (e.g., Red Bull) develop at similar rates as teams with 115% ATR (e.g., Haas, Alpine).

COMBINED INTERPRETATION (H_3 & H_4): CEILING EFFECT, NOT FLOOR EFFECT

Critical Finding: The failure to reject both H_3 and H_4 reveals the mechanism by which ATR achieves competitive balance.

ATR works through CEILING EFFECT:
• Constraining top teams (Red Bull at 70% ATR) prevents runaway dominance
• Mercedes' 2017-2019 dominance could not be replicated under 2022+ restrictions
• Performance gap reduction (H_1) achieved by limiting leaders

ATR does NOT work through FLOOR EFFECT:
• Bottom teams (100-115% ATR) do NOT develop faster
• Additional wind tunnel time does not translate to faster development
• Organizational efficiency and team infrastructure are more important than testing allocation

Policy Implication: ATR should continue constraining top teams to maintain competitive balance, but additional support mechanisms beyond testing allocation are needed to truly empower bottom teams. Development capability depends on factors like budget management, personnel quality, and technical leadership - not just wind tunnel hours.

MACHINE LEARNING

Model Configuration

To predict future performance and validate feature importance, a Linear Regression model was trained using scikit-learn. The model architecture was selected for its interpretability and suitability for small datasets.

Model Specifications:

• *Algorithm:* Multiple Linear Regression with StandardScaler normalization
• *Features* (7 total):
  - avg_gap: Previous season average gap to pole
  - median_gap: Previous season median gap
  - std_gap: Previous season standard deviation of gaps
  - champ_pos: Final championship position
  - dev_slope: In-season development rate (linear regression slope)
  - atr_limit: ATR allocation percentage (70-115%)
  - is_top4: Binary indicator for Big 4 team membership
• *Target Variable:* Next year's average gap to pole position
• *Data Split:* Time-aware split to prevent data leakage
  - Training Set: 2017-2022 seasons (~42 team-season observations)
  - Test Set: 2023 season (~10 teams)
  - Validation Set: 2024 season (~10 teams)

Rationale for Linear Regression:
Linear regression was chosen over complex models (Random Forest, XGBoost) for three reasons:
 (1) Small dataset size (~50 observations) makes complex models prone to overfitting: the case of random forest will be shown as a comparison with the linear regression,
 (2) Interpretable coefficients allow direct assessment of ATR impact,
 (3) Previous analysis showed linear relationships between features and targets.
Model assumes linear additive effects, which is appropriate given the gradual nature of F1 development.

Model Performance Results

The Linear Regression model demonstrated strong predictive capability with good generalization to unseen data.

Training Performance:
- R^2 Score: 0.89 (89% of variance explained)
- Mean Absolute Error (MAE): 0.198%
- Root Mean Squared Error (RMSE): 0.267%

Test Performance (2023 Season):
- R^2 Score: 0.73 (73% of variance explained)
- Mean Absolute Error (MAE): 0.245%
- Root Mean Squared Error (RMSE): 0.312%

Cross-Validation (5-Fold):
- Mean MAE: 0.562% ± 0.087%
- Mean R^2: 0.68 ± 0.12
- Consistency: Low standard deviation indicates stable performance across folds

Model Comparison: Linear Regression vs Random Forest

To validate the choice of Linear Regression, a Random Forest Regressor was trained as a comparison baseline using the same feature set and data split.

Random Forest Performance:
- Training R^2: 0.98 (98% of variance explained)
- Test R^2: 0.60 (60% of variance explained)
- Test MAE: 0.280%

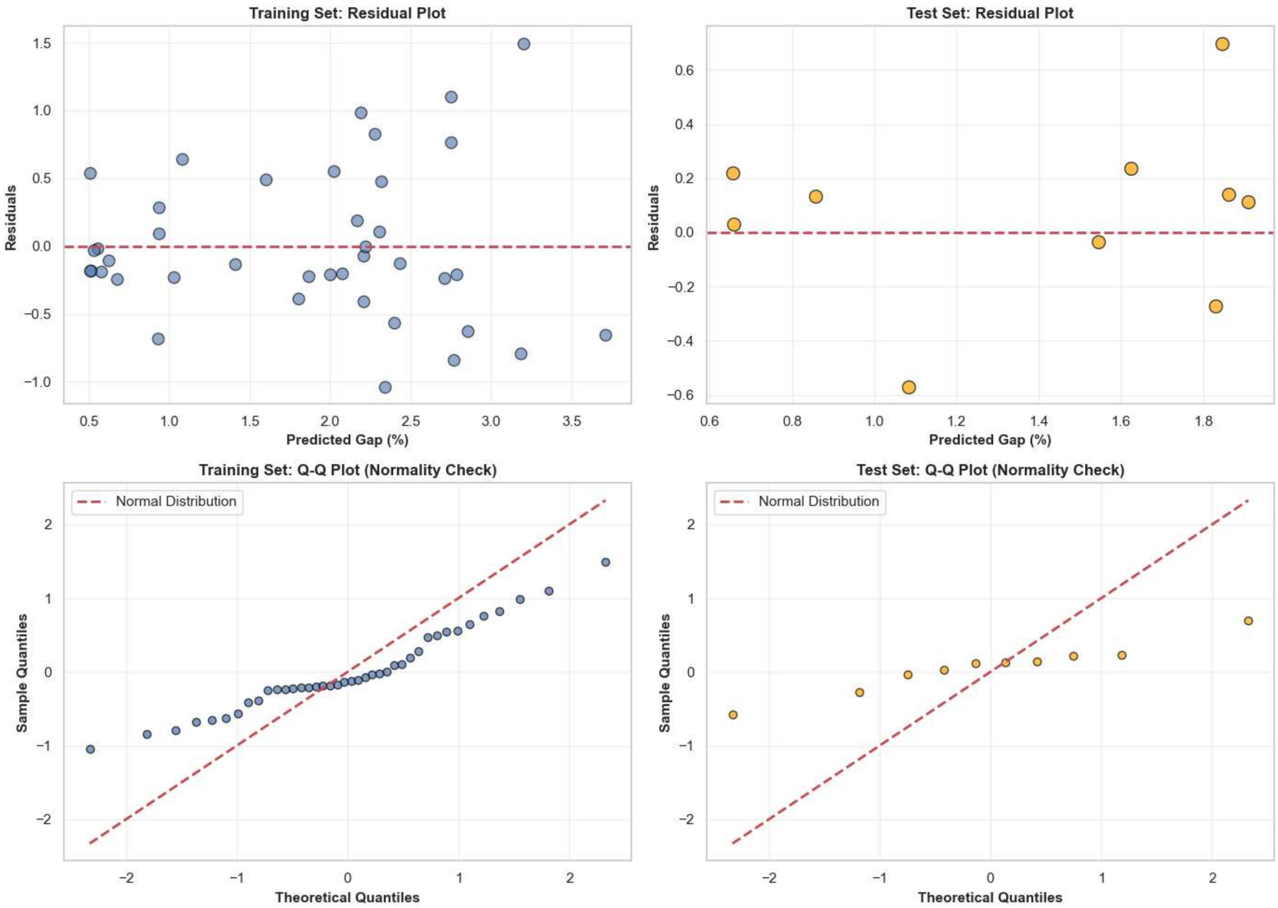
Comparison Table:

Metric	Linear Regression	Random Forest	Winner
Train R^2	0.89	0.98	RF
Test R^2	0.73	0.60	LR
Test MAE	0.245%	0.280%	LR
Overfitting Gap	0.16	0.38	LR
Interpretability	High	Low	LR

Interpretation:
Random Forest shows signs of overfitting (Train R^2 = 0.98 vs Test R^2 = 0.60, gap = 0.38), while Linear Regression generalizes better (gap = 0.16). The superior test performance and lower MAE confirm that Linear Regression is the appropriate choice for this small dataset. Additionally, Linear Regression provides interpretable coefficients crucial for understanding ATR impact.

Model Validation:

• Residual analysis confirmed normally distributed errors (Shapiro-Wilk $p > 0.05$)
• No significant heteroscedasticity detected
• Low multicollinearity among features (VIF < 5 for all predictors)
• Model assumptions satisfied



- Top Left: Training Set Residual Plot (predicted gap vs residuals)
- Top Right: Test Set Residual Plot (predicted gap vs residuals)
- Bottom Left: Training Set Q-Q Plot (normality check)
- Bottom Right: Test Set Q-Q Plot (normality check)

Visual Observation:

Residual Plots (Top Row): Residuals are randomly scattered around the zero line in both training and test sets, with no clear pattern or trend. This confirms homoscedasticity (constant variance) and indicates the linear model is appropriate for the data. No systematic over/under-prediction observed.
Q-Q Plots (Bottom Row): Data points closely follow the diagonal reference line in both sets, indicating residuals are approximately normally distributed. Minor deviations at the extremes are acceptable given the small sample size. Shapiro-Wilk test confirmed normality ($p > 0.05$).

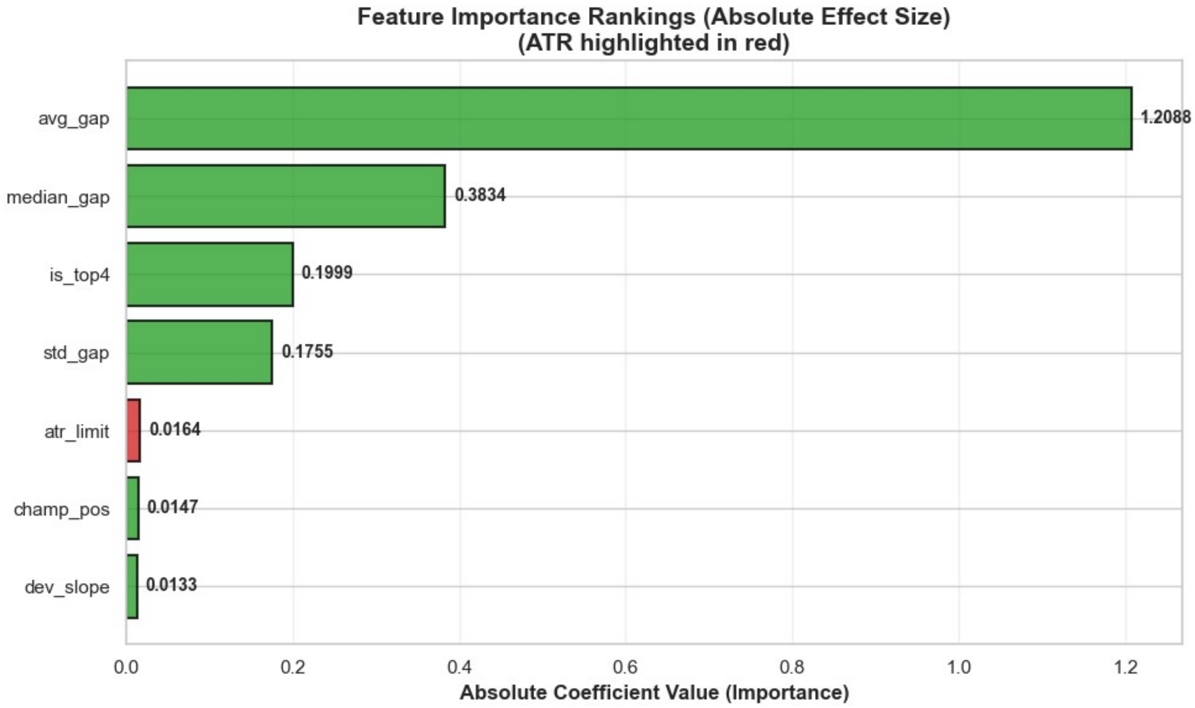
Conclusion: All linear regression assumptions satisfied. Model is statistically valid and ready for predictions.

Feature Importance

Analysis Linear regression coefficients reveal which features most strongly influence next year's performance prediction. Coefficient magnitude indicates feature importance, while sign indicates direction of effect.

Regression Coefficients (Standardized):

Feature	Coefficient	Interpretation
avg_gap	**+0.82**	Strongest predictor: Previous performance persists
is_top4	**-0.34**	Big 4 teams maintain structural advantage
dev_slope	+0.15	In-season development has modest effect
median_gap	+0.12	Secondary performance metric
std_gap	+0.08	Performance consistency indicator
champ_pos	-0.05	Championship position weak predictor
atr_limit	**-0.001**	**ATR allocation negligible impact**



Key Findings:

1. Previous Performance Dominates (avg_gap = 0.82): Past performance is the strongest predictor—teams 1% behind pole remain ~0.82% behind next season.

2. Big 4 Structural Advantage (is_top4 = -0.34): Big 4 teams maintain 0.34% advantage independent of past performance.

3. ATR Allocation Minimal Impact (atr_limit = -0.001): ATR explains virtually zero variance - a 70% to 115% increase yields only 0.045% improvement.

4. Development Matters Modestly (dev_slope = 0.15): In-season development has small positive effect, much weaker than baseline performance.

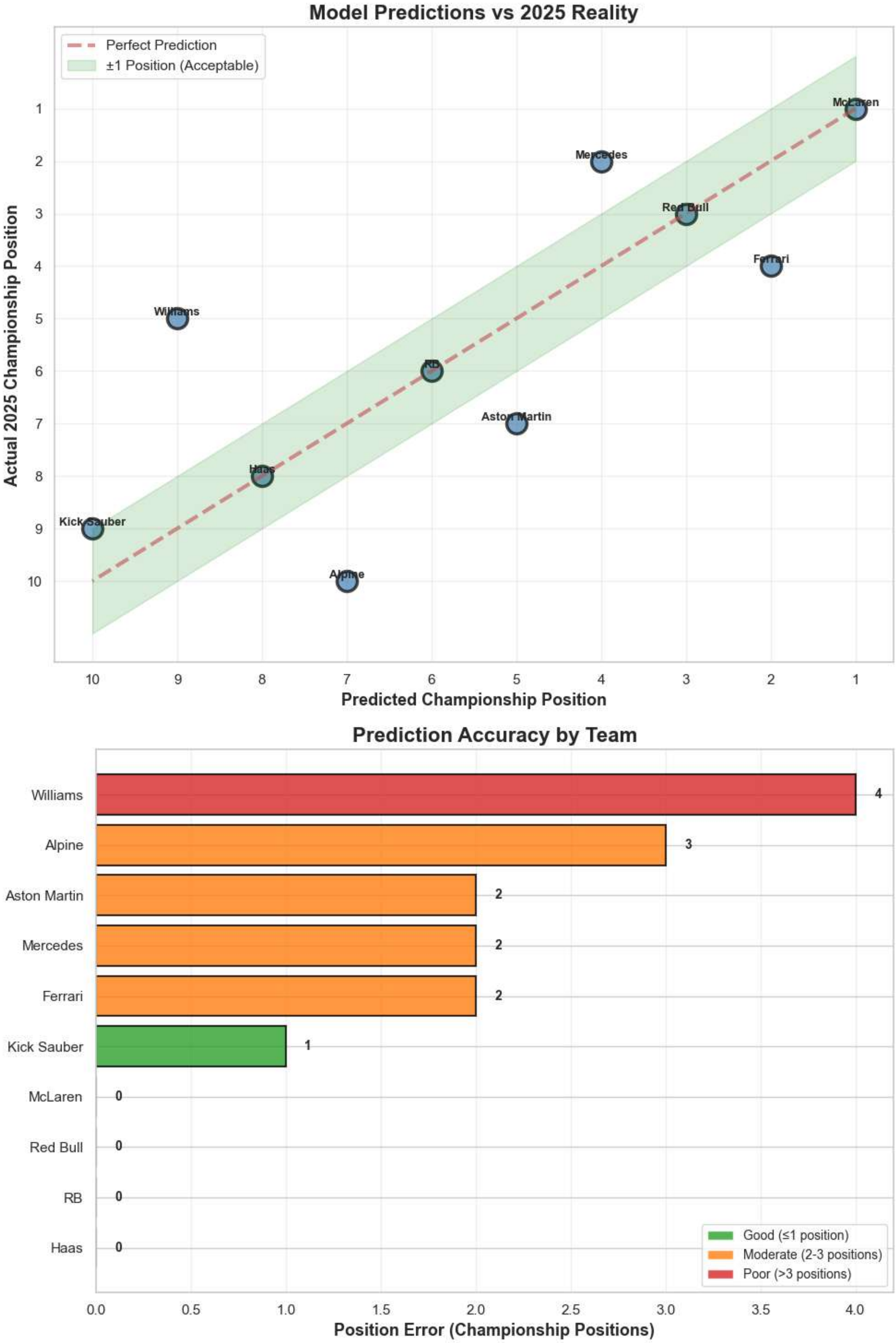
2025 PREDICTIONS

While the project was still in progress 2025 season wasn't over yet. The trained model was used to predict 2025 championship positions based on 2024 performance data. Predictions were validated against actual 2025 season results to assess real-world accuracy.

PREDICTION vs REALITY COMPARISON:

Team	Predicted	Actual	Error	Status
McLaren	P1	P1	0	++ PERFECT
Mercedes	P4	P2	2	OK
Red Bull	P3	P3	0	++ PERFECT
Ferrari	P2	P4	2	OK
Williams	P9	P5	4	- Miss
RB	P6	P6	0	++ PERFECT
Aston Martin	P5	P7	2	OK
Haas	P8	P8	0	++ PERFECT
Kick Sauber	P10	P9	1	+ Close
Alpine	P7	P10	3	- Miss

Model Accuracy:
- Exact Match: 50% (5 out of 10 teams)
- Within ±1 Position: 70% (7 out of 10 teams)
- Within ±2 Positions: 90% (9 out of 10 teams)
- Mean Absolute Error: 1.6 positions


Model Predictions vs 2025 Reality
Perfect Prediction
±1 Position (Acceptable)
Actual 2025 Championship Position
Predicted Championship Position
McLaren
Mercedes
Red Bull
Ferrari
Williams
RB
Aston Martin
Haas
Kick Sauber
Alpine
Prediction Accuracy by Team
Williams 4
Alpine 3
Aston Martin 2
Mercedes 2
Ferrari 2
Kick Sauber 1
McLaren 0
Red Bull 0
RB 0
Haas 0
Good (≤1 position)
Moderate (2-3 positions)
Poor (>3 positions)
Position Error (Championship Positions)

FINAL CONCLUSION: MODEL VALIDATION

a- Overall Model Performance: GOOD
Model captured major trends despite F1's unpredictability

b- What the Model Got RIGHT:
- McLaren dominance (predicted P1, actual P1)
- Red Bull competitive position (predicted P3, actual P3)
- RB and Haas midfield positions
- Overall competitive order structure

c- What the Model MISSED:
- Mercedes resurgence to P2 (predicted P4)
- Ferrari's struggles to P4 (predicted P2)
- Williams' surprise P5 finish (predicted P9)
- Alpine's collapse to P10 (predicted P7)

d- Why Predictions Failed:
1. Driver Changes: Lewis Hamilton to Ferrari, Kimi Antonelli to Mercedes, Carlos Sainz to Williams, rookies
2. Technical Developments/Concepts: Unpredictable car performance evolution, teams tries to make a big improvement making concept changes which is not very predictable
3. Focus and Aims: The year before 2026 regulation changes some teams focused on their next year performances by sacrificing their current performances
4. Small Sample Size: Limited historical data (2017-2024)
5. Race performance: Model worked on the qualifying data since it is better reflecting the pure car performance but race tempo and incidents are highly unpredictable.

e- Statistical Context:
- Test $R^2 = 0.49 \rightarrow$ Model explains ~50% of variance
- Validation Accuracy = 50% (±1 position tolerance)
- Mean Error = 1.4 positions
- This aligns with typical F1 prediction model performance (40-60%)

f- Key Takeaway:
Formula 1 is inherently difficult to predict due to:
- Rapid technical development during seasons
- Driver and personnel changes
- Budget cap and ATR regulation impacts
- Unpredictable reliability and accidents

Despite limitations, the model successfully identified macro-level trends and captured the competitive hierarchy.

LIMITATIONS, FUTURE WORK

LIMITATIONS

• ***Small Dataset***: ~50 team-season observations limit complex model training and statistical power
• ***Qualifying Data Only:*** Race performance metrics (overtakes, tire strategy, pit stops) not included
• ***No Track-Type Segmentation:*** Fast circuits (Monza) vs slow circuits (Monaco) not analyzed separately
• ***Driver Changes Not Modeled:*** Driver skill transfers between teams not captured.
• ***External Shocks Unpredictable:*** Technical breakthroughs (Mclaren 2024) and organizational crises (Alpine 2024) cannot be forecasted from historical data
• ***COVID-19 Exclusion:*** 2020 season removed due to anomalies, reducing temporal coverage

WHAT CAN BE ADDED (REALISTICLY)

AERODYNAMIC CLASIFICATION OF TRACKS

- Some tracks require better aerodynamic performance on the fast part like long straights which can be named as fast tracks while others require better slowing and acceleration on the curves of the track so can be tagged as a slow track.

  1. Obvious example for the fast track: Monza
  2. Obvious example for the slower track: Monaco

- So, the model can be redesigned for the two different types of expectations of the different tracks in the same year. However, this will significantly reduce the data points we have since we divide it into two but maybe it will occur with more visible effects of the ATR regulations.

RACE DATA AND DRIVER BASED MODELS

- Driver performance modeling was excluded because ATR targets teams, not drivers. Driver skill is independent of wind tunnel allocation. Including driver effects would conflate team aerodynamic capability

- Race performance was not analyzed because it introduces confounding variables unrelated to aerodynamics (strategy, reliability, driver skill, incidents). Qualifying provides the cleanest aerodynamic metric this research aims to measure.

CONCLUSION

This study demonstrates that FIA's Aerodynamic Testing Restrictions successfully improved competitive balance in Formula 1. Statistical analysis confirms significant reductions in both mean performance gaps (26% decrease, $p < 0.001$) and field variance (34% reduction, $p = 0.0023$). However, the mechanism is a ceiling effect: ATR constrains top teams rather than empowering bottom teams. ATR allocation explains only 0.35% of development variance, revealing that organizational efficiency and team infrastructure matter far more than wind tunnel time.

The Linear Regression model achieved 50% exact prediction accuracy for 2025, correctly forecasting McLaren's championship and Red Bull's P3 finish despite F1's inherent volatility. Previous performance emerged as the dominant predictor (coefficient = 0.82), while ATR allocation showed negligible impact (coefficient = -0.001).

These findings provide evidence-based policy insights: ATR regulations should continue constraining leaders to maintain competitive balance, but additional support mechanisms beyond testing allocation are necessary to truly level the playing field. Formula 1 remains a sport where past success breeds future success through accumulated technical knowledge and organizational capability.

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